In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

In [2]:
```python
df=pd.read_csv("house_price.csv")
df.head()
```

Out[2]:

	House Size	Bedrooms	Age of House	House Price
0	1500	3	10	350000
1	1800	3	8	400000
2	2000	4	5	450000
3	2200	4	2	475000
4	2500	5	15	525000

In [3]:
```python
x = df[['House Size', 'Bedrooms', 'Age of House']]
y = df['House Price']
```

In [4]:
```python
X_train,X_test,Y_train,Y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

In [5]:
```python
model=LinearRegression()
model.fit(X_train,Y_train)
```

Out[5]:

▾ LinearRegression

▶ Parameters

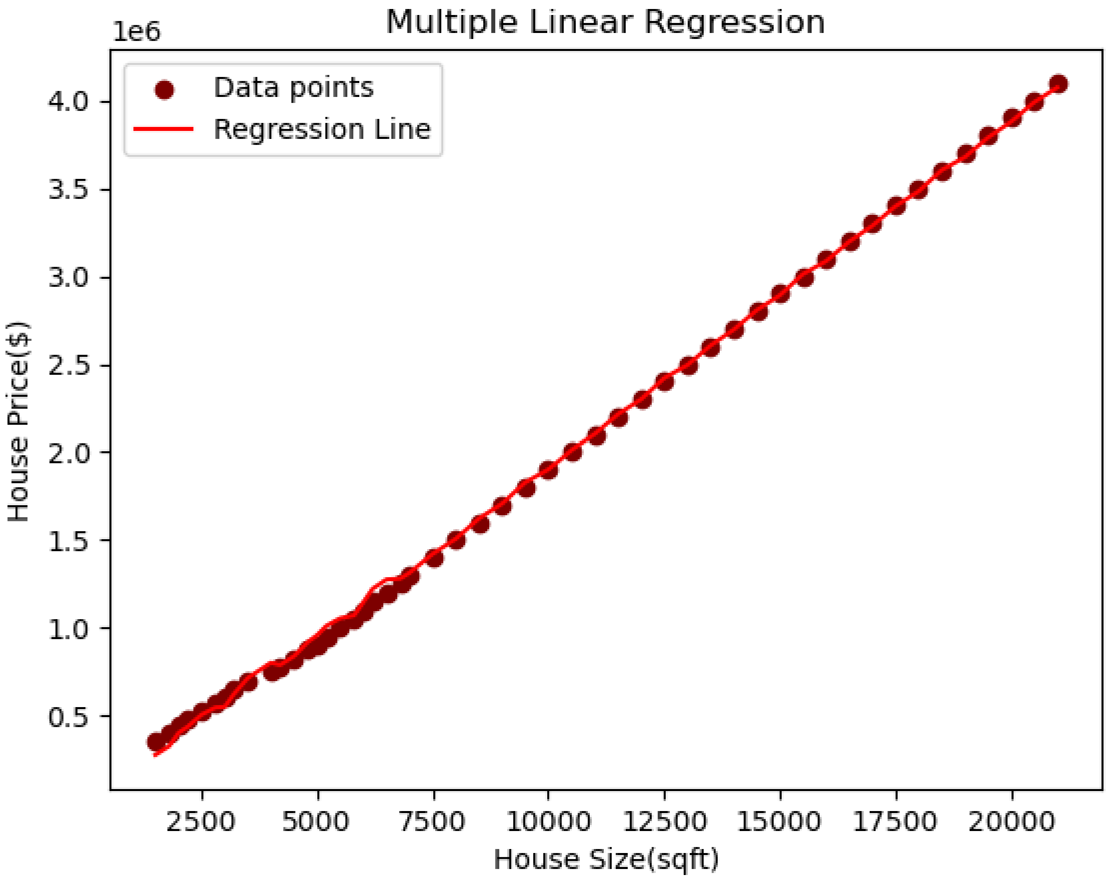
In [6]:
```python
Y_pred=model.predict(X_test)
Y_pred
```

Out[6]:
```
array([ 958198.91326438, 3084757.95178209, 2213923.0591368 ,
        3676539.75229267, 1145827.56632132, 3989268.46066852,
        1827657.81346492, 1706685.73548026, 2419439.6139755 ,
        1275055.59897449])
```

In [7]:
```
r2_score=r2_score(Y_test,Y_pred)
r2_score
#print evaluation metrics
print(f"R-squared Score:",r2_score)
```

```
R-squared Score: 0.9987416757120748
```

In [8]:
```
#plot
plt.scatter(x['House Size'],y,color='maroon',label='Data points')  #All data points
plt.plot(x['House Size'],model.predict(x),color='red',label='Regression Line') #Regression line
plt.xlabel('House Size(sqft)')
plt.ylabel('House Price($)')
plt.title('Multiple Linear Regression')
plt.legend()
plt.show()
```



Multiple Linear Regression
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
House Price($)
Data points
Regression Line
2500
5000
7500
10000
12500
15000
17500
20000
House Size(sqft)